

Moment Equations

Tom De Fraja

11/11/22

Boltzmann Equation to Moment equations

$$\frac{\partial f}{\partial t} + c_k \frac{\partial f}{\partial x_k} + G_k \frac{\partial f}{\partial c_k} = S$$

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$$\psi \cdot \left(\frac{\partial f}{\partial t} + c_k \frac{\partial f}{\partial x_k} + G_k \frac{\partial f}{\partial c_k} = S \right)$$

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$$m \int \psi \cdot \left(\frac{\partial f}{\partial t} + c_k \frac{\partial f}{\partial x_k} + G_k \frac{\partial f}{\partial c_k} = S \right) d\mathbf{c}$$

Boltzmann Equation to Moment equations

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The evolution of the moment $u_\psi = m \int \psi f d\mathbf{c} =: \langle \psi f \rangle$ is

$$\frac{\partial \langle \psi f \rangle}{\partial t} + \frac{\partial \langle c_k \psi f \rangle}{\partial x_k} = G_k \left\langle \frac{\partial \psi}{\partial c_k} f \right\rangle + S_\psi$$

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Conservation equations ($S_\psi = 0$)

$$\psi = 1 \quad \rightsquigarrow \quad \frac{\partial \rho}{\partial t} + \frac{\partial \rho v_k}{\partial x_k} = 0$$

$$\psi = c_i \quad \rightsquigarrow \quad \frac{\partial \rho v_i}{\partial t} + \frac{\partial \rho \theta}{\partial x_k} + \frac{\partial \sigma_{ik}}{\partial x_k} = \rho G_i$$

$$\psi = \frac{1}{2} c^2 \quad \rightsquigarrow \quad \frac{\partial \rho e}{\partial t} + \frac{\partial \rho e v_k + p_{ik} v_i}{\partial x_k} + \frac{\partial q_k}{\partial x_k} = \rho G_k v_k$$

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$$\psi = \frac{1}{2} C^2 \quad \rightsquigarrow \quad \rho \frac{\partial \theta}{\partial t} + \rho v_k \frac{\partial \theta}{\partial x_k} + \frac{\partial q_k}{\partial x_k} = -p_{ij} \frac{\partial v_k}{\partial x_k}$$

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$$\frac{D\sigma_{ij}}{Dt} + \frac{\partial u_{ijk}^0}{\partial x_k} + \frac{4}{5} \frac{\partial q_{\langle i}}{\partial x_{j \rangle}} + 2\sigma_{k\langle i} \frac{\partial v_{j \rangle}}{\partial x_k} + \sigma_{ij} \frac{\partial v_k}{\partial x_k} + 2\rho\theta \frac{\partial v_{\langle i}}{\partial x_{j \rangle}} = -\frac{p}{\mu} \sigma_{ij}$$

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With $\psi = C^2 C_i$, corresponding to the heat flux variable q_i , we get

$$\begin{aligned} \frac{Dq_i}{Dt} + \frac{5}{2} \rho\theta \frac{\partial \theta}{\partial x_i} - \sigma_{ik} \frac{\partial \theta}{\partial x_k} - \sigma_{ik} \theta \frac{\partial \ln \rho}{\partial x_k} - \frac{\sigma_{ik}}{\rho} \frac{\partial \sigma_{kl}}{\partial x_l} - \frac{5}{2} \theta \frac{\partial \sigma_{ik}}{\partial x_k} + \frac{1}{2} \frac{\partial u_{ik}^1}{\partial x_k} \\ + \frac{1}{6} \frac{\partial w^2}{\partial x_i} + u_{ikl}^0 \frac{\partial v_k}{\partial x_l} + \frac{7}{5} q_i \frac{\partial v_k}{\partial x_k} + \frac{7}{5} q_k \frac{\partial v_i}{\partial x_k} + \frac{2}{5} q_k \frac{\partial v_k}{\partial x_i} = \frac{2}{3} \frac{p}{\mu} q_i \end{aligned}$$

Linearised moment equations

Conservation equations:

$$\begin{aligned}\frac{\partial \rho}{\partial t} + \frac{\partial v_k}{\partial x_k} &= 0 \\ \frac{\partial v_i}{\partial t} + \frac{\partial \rho}{\partial x_i} + \frac{\partial \theta}{\partial x_i} + \frac{\partial \sigma_{ik}}{\partial x_k} &= G_i \\ \frac{\partial \theta}{\partial t} + \frac{\partial q_k}{\partial x_k} + \frac{\partial v_k}{\partial x_k} &= 0\end{aligned}$$

Higher moments:

$$\begin{aligned}\frac{\partial \sigma_{ij}}{\partial t} + \frac{4}{5} \frac{\partial q_{\langle i}}{\partial x_{j \rangle}} + 2 \frac{\partial v_{\langle i}}{\partial x_{j \rangle}} + \frac{\partial m_{ijk}}{\partial x_k} &= -\frac{\sigma_{ij}}{\text{Kn}}, \\ \frac{\partial q_i}{\partial t} + \frac{5}{2} \frac{\partial \theta}{\partial x_i} + \frac{\partial \sigma_{ik}}{\partial x_k} + \frac{1}{2} \frac{\partial R_{ik}}{\partial x_k} + \frac{1}{6} \frac{\partial \Delta}{\partial x_i} &= -\frac{2}{3} \frac{q_i}{\text{Kn}}.\end{aligned}$$

Closure

Recall the evolution of the moment $u_\psi = \int \psi f d\mathbf{c} =: \langle \psi f \rangle$ from the Boltzmann equation

$$\frac{\partial \rho \langle \psi f \rangle}{\partial t} + \frac{\partial \rho \langle c_k \psi f \rangle}{\partial x_k} = \rho \left\langle G_k \frac{\partial \psi}{\partial c_k} \right\rangle + S_\psi$$

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On top of the variable u_ψ , this involves the variables $u_{c_k \psi}$ and $u_{\partial_k \psi}$, the “n+1”- and “n-1”-th moments.

Must find a way to close the system.

Navier-Stokes

To derive Navier-Stokes, considered the moments $\{1, c_i, C^2\}$,
hence the variables $\{\rho, \rho v_i, \rho \theta\}$:

Conservation equations:

$$\frac{\partial \rho}{\partial t} + \frac{\partial v_k}{\partial x_k} = 0$$
$$\frac{\partial v_i}{\partial t} + \frac{\partial \rho}{\partial x_i} + \frac{\partial \theta}{\partial x_i} + \frac{\partial \sigma_{ik}}{\partial x_k} = G_i$$
$$\frac{\partial \theta}{\partial t} + \frac{\partial q_k}{\partial x_k} + \frac{\partial v_k}{\partial x_k} = 0$$

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Need expressions for σ_{ij} and q_i .

Euler

$$\sigma_{ij} = 0, \quad q_i = 0;$$

Navier-Stokes

$$\sigma_{ij} = -2\text{Kn} \frac{\partial v_{\langle i}}{\partial x_{j\rangle}}, \quad q_i = -\frac{5}{2} \frac{\text{Kn}}{\text{Pr}} \frac{\partial \theta}{\partial x_i}$$

13-moment equations

The next set we derived we considered the moments

$\{1, c_i, C^2, C_i C_j, C^2 C_i\}$, hence the variables $\{\rho, \rho v_i, \rho \theta, \sigma_{ij}, q_i\}$:

Higher moments:

$$\frac{\partial \sigma_{il}}{\partial t} + \frac{4}{5} \frac{\partial q_{\langle i}}{\partial x_{l \rangle}} + 2 \frac{\partial v_{\langle i}}{\partial x_{j \rangle}} + \frac{\partial m_{ilk}}{\partial x_k} = - \frac{\sigma_{il}}{\text{Kn}},$$
$$\frac{\partial q_i}{\partial t} + \frac{5}{2} \frac{\partial \theta}{\partial x_i} + \frac{\partial \sigma_{ik}}{\partial x_k} + \frac{1}{2} \frac{\partial R_{ik}}{\partial x_k} + \frac{1}{6} \frac{\partial \Delta}{\partial x_i} = - \frac{2}{3} \frac{q_i}{\text{Kn}}.$$

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Derivations due to Harold Grad, and Struchtrup and Torrilhon:

Grad 13

R13

$$R_{ij} = 0, m_{ilk} = 0, \Delta = 0;$$
$$R_{ij} = -\frac{24}{5} \text{Kn} \frac{\partial q_{\langle i}}{\partial x_{j \rangle}}, m_{ilk} = -2 \text{Kn} \frac{\partial \sigma_{\langle il}}{\partial x_{k \rangle}},$$
$$\Delta = -12 \text{Kn} \frac{\partial q_k}{\partial x_k}$$

Moment Equations

Name	Kn order	Variables	Higher terms	Year
Euler	0th	ρ, v_i, θ	$\sigma_{ij} = q_i = 0$	1750s
Navier-Stokes	1st	ρ, v_i, θ	$\sigma_{ij} = -2Kn\partial_{\langle i}v_{j\rangle}$ $q_i = -cKn\partial_i\theta$	1840s
Grad 13	2nd	$\rho, v_i, \theta,$ σ_{ij}, q_i	$R_{ij} = m_{ilk} = \Delta = 0$	1958
R13	3rd	$\rho, v_i, \theta,$ σ_{ij}, q_i	$R_{ij} = -\frac{24}{5}Kn\partial_{\langle i}q_{j\rangle}$ $m_{ilk} = -2Kn\partial_{\langle i}\sigma_{lk\rangle}$ $\Delta = -12Kn\partial_k q_k$	2003

Test

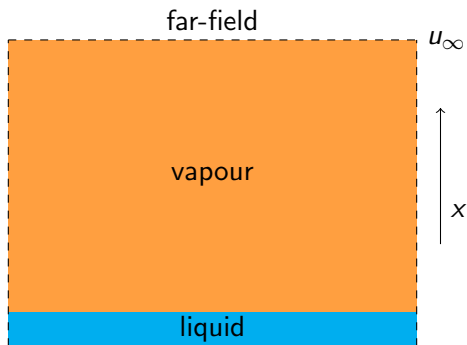
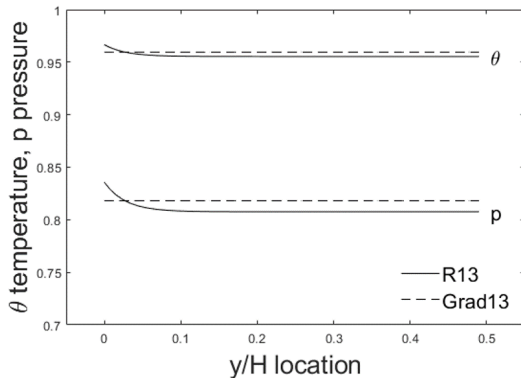


Figure: Set-up for half-space evaporation, with velocity prescribed at the far field.

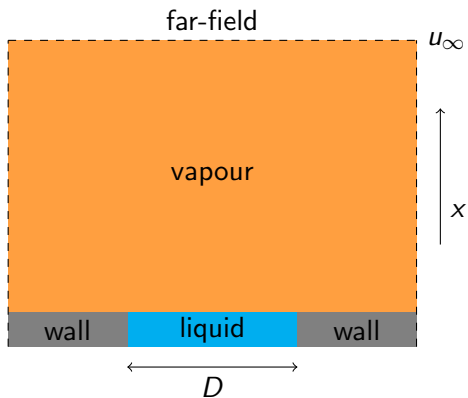
Result



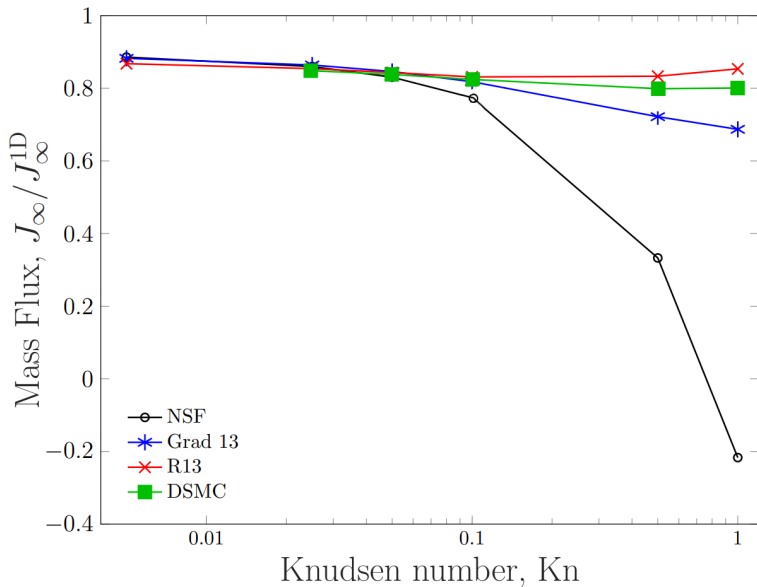
due to appearance of equation $\sigma \propto \text{Kn}^2 \sigma''$

$$\Rightarrow \rho \propto \rho_{\infty} - A \exp(-cx/\text{Kn})$$

Test 2



Mass flux vs Knudsen number



Fin

Thanks!